

Installation & Application Instructions for Premise Tight Buffered Cable



NOTE:

Except as may be otherwise provided by contract, these drawings and/or specifications are the property of AFL Telecommunications, are issued in strict confidence, and shall not be reproduced or copied or used as the basis for manufacture or sale of product without permission.

Certain information such as the data, opinions or recommendations set forth herein or given by AFL Representatives, is intended as a general guide only. Each installation of overhead electrical conductor, underground electrical conductor, and/or conductor accessories involves special conditions creating problems that require individual solutions and, therefore, the recipient of this information has the sole responsibility in connection with the use of the information. AFL does not assume any liability in connection with such information.

TABLE OF CONTENTS

CONTENTS	PAGE
General Information 1.0	3
Safety 2.0	3
Pre-Installation Procedure 3.0	3
Applications 4.0	4
Cable Handling 5.0	4
Storage and Handling 6.0	6
Cable Preparation 7.0	6
Construction Layers 8.0	7
Generic Cabling System 9.0	8
Centralised Cross Connect 10.0	8
Cable Installation Techniques 11.0	9
Reinstatement of Site on Completion 12.0	11
Documentation 13.0	11

1.0 GENERAL INFORMATION

The following instructions provide general information for engineers, consultants, project managers, and construction personnel on the suitable methods for installing Premise fibre optic cables. Installation equipment mentioned in this document must comply with equipment manufacturers' recommendations and regulations. Please contact AFL with questions about any method or equipment recommendations.

2.0 SAFETY

AFL strongly recommends the use of approved personal protection equipment as needed per installation. All protection equipment must meet the relevant safety requirements, and all personnel involved should be aware of all applicable Safe Work Australia regulations, state and local regulations, and company policies.

2.1 WARNING

Beware of optical power exposure from light sources such as LED, VCSEL / LASER, or interfaces like optical fibre fragments. Never look directly into the end of the fibre optic cables until it is confirmed that there is no light source at the other end. There is a possibility of optical fibre fragments piercing into the skin/eyes during splicing or installation process. This might lead to infection or difficulty in removing.

Check for any toxic chemicals or fumes that can cause allergic reactions while preparing the optical fibres. Precaution needs to be taken while drilling or working closely with products containing asbestos during installation. Beware of Electric shock and Earth Potential Rise while installing near High-Voltage zones. Increased voltage in customer premises also has the capacity to damage test equipment and cablers should take reasonable precautions to prevent this from happening. Precautions must be also while installing cables with metallic elements present in it.

3.0 PRE-INSTALLATION PROCEDURE

Make sure Before You Dig Australia (BYDA) plans and the client's construction plans are on site, and all underground utilities are physically located. Identify areas that may require special consideration to install cables. Risk assessments must be carried out as a part of risk management. Machinery, equipment, and tools used for installation must be examined for proper condition of use during this time.

Use Job Safety Analysis or similar safety management tools to eliminate and/or control the hazards. Keep the time between the opening of the ground surfaces and the final restoration as short as practically possible. Visually inspect the cable reels for any transportation damage. Ensure that the manufacturers cable storage and handling guide is followed.

4.0 APPLICATIONS

- Indoor/Outdoor Tight Buffered cables are specified for campus network cabling between buildings where interbuilding lengths are short enough.
- Circular Premise cable based on 900µm tight buffer constructions provide the performance and density demanded by today's optical network installations.
- Cable designs support either direct or indirect termination schemes while enabling efficient routing through industry standard Fibre Management Systems.
- Cable constructions support the application of Industry leading multi-mode fibre as well as singlemode solutions.
- Installers can recognize savings from the lower costs of terminating tight buffered cables.
- It is constructed completely of LSZH materials, PVC-free.
- LSZH jacket material makes the cable flame retardant as is self-extinguishing, meeting the local standard requirements.
- No toxic fumes or gases are released when these cables are exposed to fire and contains smoke emission during burn.
- The cable has 'Dry' water blocking properties; therefore, suitable for protected use in external covered tray or free draining conduit.

Used in:

- Multi-dwelling units
- Building interconnections
- Campus LAN
- Datacentre trunk cabling
- Pre-terminated optical assembly
- Optical fibre cable distribution

5.0 CABLE HANDLING

5.1 TENSION

Before the cable is pulled through for installation, it is necessary to know the tensile strength of the cable to prevent breakage of fibres and maintain long-term reliability. Exceeding the cable's maximum installation tension, can damage the cables' mechanical and optical performance, void the warranty, or potentially break a fibre. When pulling or securing the cable, it is important to use the strength element(s) and not the cable jacket and to monitor the pull tension on the cable continuously using proper equipment. Always keep cable pulling tension in check and caution not to exceed the value at any time throughout the process. Please refer AFL's cable technical datasheets.



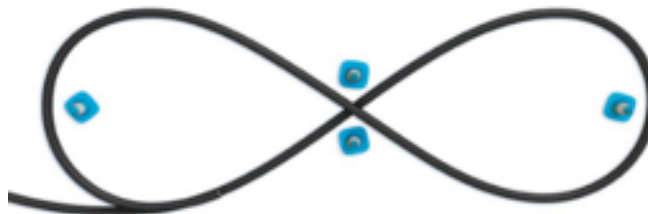
5.2 MINIMUM BEND RADIUS

Refer AFL Cable technical datasheets for minimum bending radius values during installation (Under load) and after installation (No load). These values are determined by AFL, to be the smallest bend that the cable can withstand before any increase in attenuation.



5.3 KINK

Cables should never be kinked or knotted. It is important to especially monitor the cable reels during unwinding, to ensure that no mechanical damage such as kinks and/or twists occurs. Always roll the cable off the drum while de-reeling instead of spinning it off the spool (flange) end, as this will put a twist in the cable for every turn on the spool. During excess slack at pulling locations, it is recommended to lay the cable in a figure "8" to prevent cable twist. Kinking/twisting the cable damages the outer layers and places strain on the fibres, which may lead to high attenuation or even fibre breaks.



5.4 CRUSH / IMPACT

The crush performance of a cable determines the ability of the cable to not only withstand load but also recover once the load has been removed. Cables can be subjected to crush and impact under different circumstances ranging from cable ties used for installation, a hammer dropped on the cable, while hauling in duct or a vehicle driven over the cable by mistake. Care must always be taken before and during installation to not exceed the crush and impact values provided in AFL's cable technical datasheet.

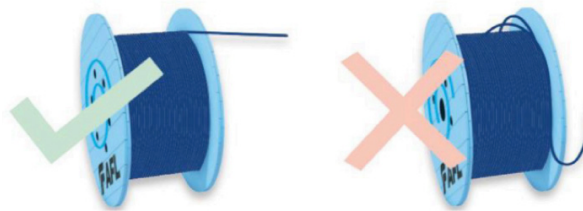


6.0 STORAGE AND HANDLING

Cable reels must always stand upright with flanges placed vertically on a flat surface. Leave enough space between stored drums for air circulation and chocks/timber blocks must be placed on either side of the flanges to prevent it from rolling. During storage, cable ends should never be left exposed.

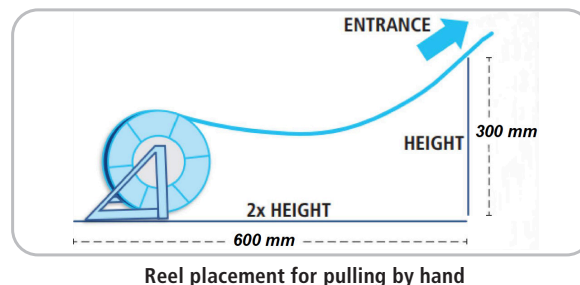
Ensure the excess length on the inner end is safely managed. When de-reeling the cable, ensure that the inner end of the cable is released from all the attachments to the flange. Decrease the pulling tension if the cable does not payoff smoothly.

All AFL cables are optically tested before they leave the manufacturing facility. Test Certificates and Certificate of Conformity by can be provided upon request. Refer "AFL Premise - Optical Characteristics" for relevant fibre type of any cable design manufactured at the AFL Australia facility. Prior to installation, physical inspection and optical testing may be required based on contractors' procedure or customer requirements.



6.1 GENERAL STEPS FOR PULLING CABLE OFF THE REEL

Step 1 Place the reel in a safe spot approximately 2x the distance of the install height.



Step 2 One team member should spool off the cable guiding it toward the pathway.

Step 3 Use a chain of team members to pass the cable from person-to-person, keeping slight slack between installers.

Step 4 Repeat Steps 2 and 3 until enough slack has been delivered to the landing side. If tension is felt in any process, stop and find the binding problem before resuming. When placing cable, maintain bend radius during the entire process.

NOTE: It may be beneficial to attach a pull line to the cable.

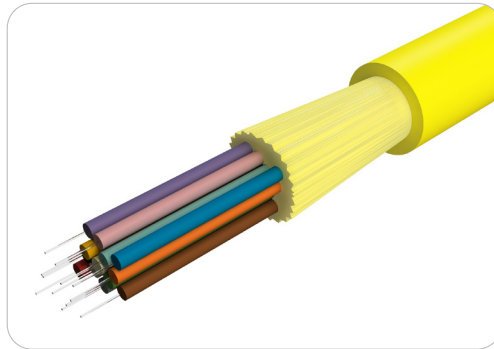
7.0 CABLE PREPARATION

Safety measures must be taken before commencing the cable / fibre stripping procedure. Ensure suitable protective personal equipment and follow local relevant safety procedures. Appropriate stripping tools must be used for easy access through the sheath layers. Try not to damage the fibres during this process.

Strength elements such as glass/ fibre rods and aramids wherever used must be cut using appropriate tools to clean and neat stripping. Refer relevant AFL's Cable Stripping guide for a comprehensive step-by-step cable stripping instruction. Fibre ends must be properly cleaved to provide satisfactory visual and geometrical end-face parameters. Use fibre end inspection tools to ensure fibres and connectors are free of debris. Ensure the tools used for cable preparation are calibrated and of appropriate size to the cable.

8.0 CONSTRUCTION LAYERS

This section details multiple elements in the cable construction that must be taken into consideration during installation.



8.1 TIGHT BUFFERED FIBRES

Acrylate coated fibre strands are buffered by extruding a thermoplastic Low Smoke Zero Halogen material. This increases the Tight Buffered fibre diameter to 900 microns, an industry standard providing the fibre with additional mechanical and environmental protection. Buffered fibres are colour coded for identification and the increased size makes fibre handling easier during installation. The buffer material is stripped back to an exact length as required while installing connectors.

8.2 CABLE STRENGTH ELEMENTS

Installation techniques and varying applications place different demands on the cable. The primary design of a cable is to package the fibre in a way that minimises/eliminates the damage and stress on the fibre itself. Core to this is the use of strength elements in the cable.

AFL Premise cables are designed with a Fibre / Glass Reinforced Plastic (FRP / GRP) as a central strength member that protects the fibres from stress by absorbing the associated load. Aramids are utilised to provide the required strength to bear the load and to retain the flexible characteristics of the cable based on cable applications.

8.3 LSZH OUTER SHEATH

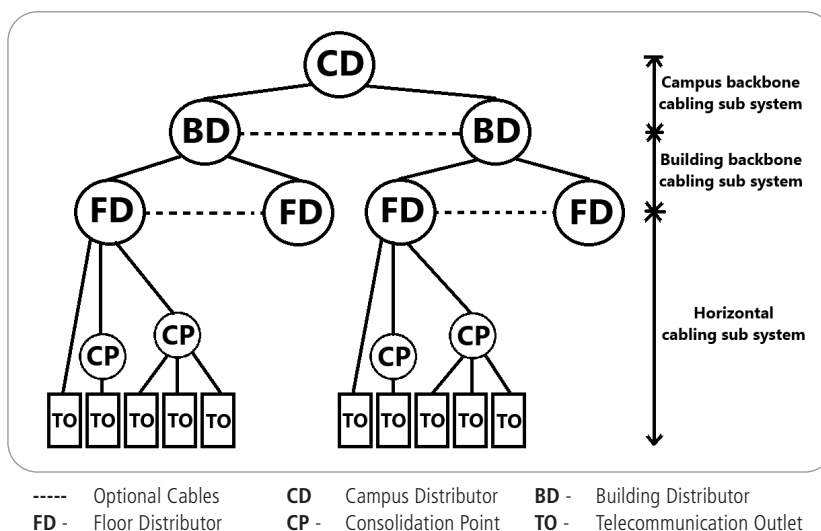
AFL premise cables are PVC free and constructed with Low Smoke Zero Halogen (LSZH) outer sheath which limits smoke emission rate when burnt and contain no toxic halogens. LSZH doesn't release dense black smoke or toxic fumes when exposed to fire. It minimises the flammability content of the cable and makes it suitable for use in indoor, free draining duct/conduit, protected (non-UV exposed) environments.

8.4 UV STABILISED CABLE

Exposure to sun's UV rays, will make some sheathing materials discolour over time and go brittle. It is significant to differentiate and understand whether the cables are going to be exposed to UV rays for a long-term or short-term. All AFL Premise cable jacket colours are UV stable however not suitable for prolonged direct UV exposure. For best performance AFL recommends a Black Outer Jacket as an outer most cable jacket in Premise cables. Cable exposure to UV only during storage and transport prior to installation is considered a short-term. AFL's Application Engineers can be a resource for providing more information regarding UV stabilisation of the cable to be purchased, based on the cable application.

9.0 GENERIC CABLING SYSTEM

The carrier network cabling extends up to the network boundary point (NBP) beyond which is the customer premises cabling. Customer premises cabling is the responsibility of the premises owner, and any work performed on this cabling will require the services of an Australian Communications and Media Authority (ACMA) Registered Cabler.



10.0 CENTRALISED CROSS CONNECT

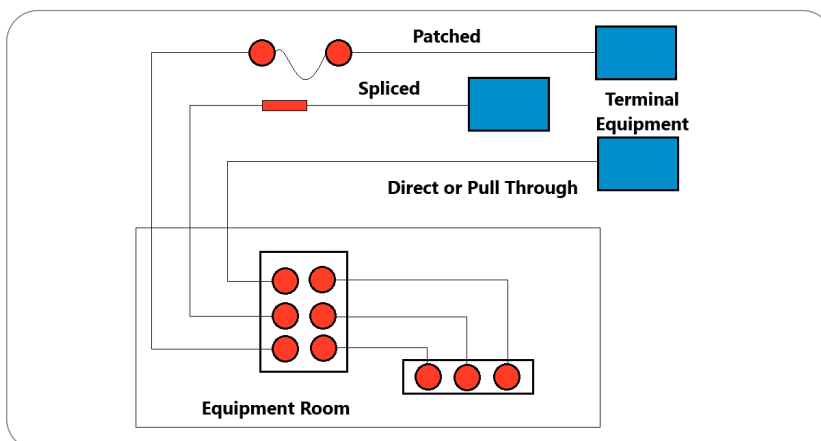
A common method of deploying fibre cables in an enterprise network for stand alone buildings or a campus environment is the Centralised Cross Connect system. In this design all fibre channels lead back to the equipment or server room but do not limit flexibility due to the ability to patch each link as required.

Centralised cross-connects are recommended by the standardisation bodies as being the preferred way to structure a fibre optic cabling system.

Some advantages of a cross connect system include:

- Modularity & Scalability
- Fast Accessibility to fibres for MAC's
- Singlemode and multi-mode integration

Some cabling systems are designed for direct link connection to equipment without the use of patch cords. This limits system flexibility but improves overall performance by reducing connector losses and reflections over the link.

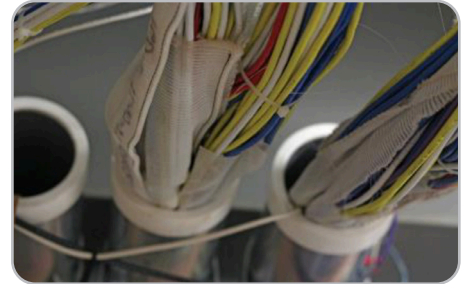


11.0 CABLE INSTALLATION TECHNIQUES

The following section provide information for installing fibre optic premise cables in various environments. Follow all local, state, and federal rules and regulations and safety guidelines.

11.1 HAND PULLING IN A DUCT / CONDUIT

- Step 1** From the landing zone, push (rod), suck a parachute or blow a pull line to the opposite side where the cable reel is located.
- Step 2** Attach a pulling grip to the cable(s) being installed.
- Step 3** Verify cable reel is placed in the proper location (approximately 2x the distance than the height of the install entrance) and is in line with duct/conduit.
- Step 4** Slowly feed the cable into the duct/conduit ensuring all sharp edges are covered, and the bend radius for the cable is not violated.
- NOTE:** Install protective plastic covers over the edges of conduit to shield the cable from damage.
- Step 5** From the landing side, pull the pull line making sure to monitor tension (one person averages over 110 N of force with one arm).
- Step 6** As the landing side is pulling, the supply side should be rolling cable off the reel while keeping slack in the line. This allows all tension in the system to be focused on pulling through the duct/conduit not split between rolling the slack off the reel.
- Step 7** Continue to pull the cable and monitor slack, sheath markings, and tension. (If, at any point, excess force is needed, STOP and check sheath markings to determine where the cable is. Then track that point down in the pathway and verify there are no issues before beginning again.)
- Step 8** When the cable is approximately 3m from the landing side, slow the pull so that excess slack is not pulled through when the cable exits the duct/conduit.
- Step 9** Once the cable exits the duct/conduit, slowly pull the cable enough to route the proper slack plus 1m.
- Step 10** Cut the first 1m of the cable to avoid the potential damaged area during installation.
- Step 11** Measure and cut the proper slack length on the launch side.



Plastic covers over conduit edges



Maintain proper slack off the reel

Hand Pulling Notes

If the install path is not continuous, then the duct/conduit needs to be connected straight through or the break the install into sections so that the cable is inside the duct/conduit from launch to land.

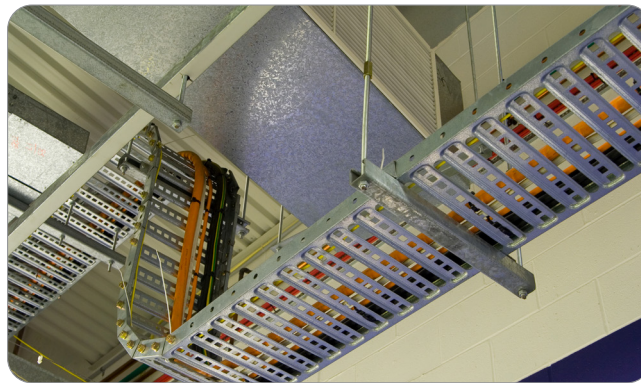
11.2 HORIZONTAL INSTALLATION

For horizontal installation of cables with low outlet density, conduit systems can be used only when workstation outlet locations are permanent and there is no significant need for wiring flexibility. This is because conduits in the floor are often concreted; hence, any future changes to the installation becomes difficult.

Ensure that the pulling tension of individual cable is not exceeded at any point during the installation. Refer to premise technical data sheet for installation tension limits. Pull boxes can be installed for fishing the run and looping the cable for next length of conduit. Inner duct is an excellent tool for protecting cable and easing future installations.

11.3 DROPPED CEILINGS AND RAISED FLOORS

Most new buildings have dropped ceilings and raised floors designed with panels that can be easily removed or opened for fast access. Suspended ceilings consist of low-weight panels supported by a system of metal frames or grids that are attached to the ceiling using struts or wires. Electronic and computer server rooms will have raised floors. Cables in these spaces should be supported and organised in trays, wireways or racks for easy maintenance. I-hooks and bridle rings can be used for support.



11.4 CABLE IN TRAYS

Cable trays or ladder racks provide a convenient, safe and efficient location in which to install optical-fibre cable. Trays can be installed in ceilings, below floors and in riser shafts. Installation usually precedes pulling the cable, because trays can be used for many other types of cable. These routes can be used for installing new cable if they run to appropriate locations.

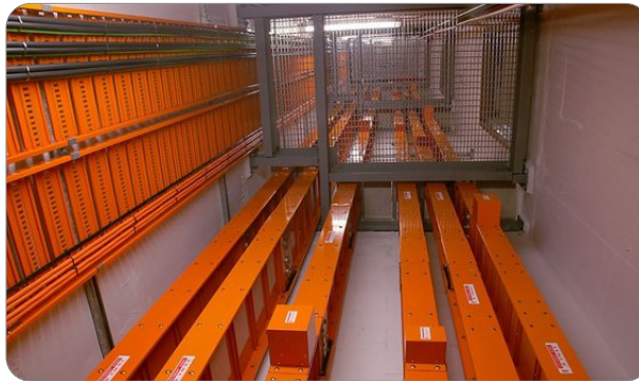
Although a tray provides support and basic protection, care should be taken to avoid as much tension, crushing and bending as possible. Routes should be inspected for sharp turns, snags (sometimes from other cables) and rough surfaces. Try to run the fibre cable without pulling it under or between heavier cable or multiple cables that may stress the fibre.



11.5 RISER INSTALLATION

Optical-fiber cables intended for vertical applications have a calculated maximum vertical rise value assigned to them. The vertical rise is the distance the cable may be pulled vertically before being supported. It is determined by the weight of the cable and its ability to resist buckling or kinking.

To pull cable vertically, use split wire mesh grips. The device works like basket or finger grips, supporting the cable without crushing the core. Cables should be supported by cable ties, straps or clamps in wiring closets. Whenever possible, begin the installation from the top, allowing the weight of the cable to help the pull rather than adding more load.



12.0 REINSTATEMENT OF SITE ON COMPLETION

Ensure that all excavation works are done to high standard with minimum disruption to the surrounding environment as much as possible. Any results that show damage or degraded characteristics of the cable under test must be rectified as per standard procedures. In any case of damage made to the property or assets in the job site, customers must be notified, and appropriate repair or alterations must be made. On completion of the work remove all plant, materials and other things that may have been brought upon the site in aid of works. Try to leave the site in a similar or better condition to how it was before the commencement of job. Ensure final sign-off on the work is received from the customer.

13.0 DOCUMENTATION

It is good practice to complete a detailed As-Built drawing and update as necessary after maintenance or repairs. Record all changes to designs and installation arrangements and forward to all stakeholders. AFL's Application Engineers can be a resource for providing information regarding previously purchased cable. Cable test reports must be part of the work completion reports. Detailed record keeping and documentation often are beneficial for future reference. Follow customer requirements and adhere to local, state, federal and industry codes, and regulations.